

Integrated Civic Reporting & Legislative Engagement Platform

1. Problem Statement

Currently, citizens in Namibia, particularly within Windhoek, lack a single, consolidated platform to report municipal infrastructure failures (such as potholes, water leaks, power outages, or uncollected garbage), lodge direct feedback with government ministries, comment on proposed legislation, or directly submit formal suggestions to specific ministries.

Residents are forced to download, register for, and navigate multiple separate applications, websites, and unstructured communication channels.

This disjointed landscape creates significant cognitive overload, confusion, and frustration for citizens, who are left uncertain about which channel belongs to which specific service or municipal department. The obligation to manage multiple single-purpose apps for routine civic duties creates high friction, leading many residents to abandon their attempts to report critical issues altogether. Consequently, public trust in municipal responsiveness erodes, while municipal and national authorities suffer from siloed data, duplicate reporting, and an inability to maintain a single, integrated view of public needs.

The existing process is fragmented across disjointed channels, leading to slow response times, limited citizen participation in public governance, and a lack of transparency.

2. Affected Parties

- **Namibian Citizens (specifically Windhoek residents):** Who experience overwhelming app fatigue, confusion, and inconvenience from having to juggle multiple single-purpose digital tools or accounts just to access basic public services, lodge complaints, or participate in democratic decision-making. This lack of a unified touchpoint causes frustration and leads many citizens to disengage and leave vital public issues unreported.
- **Government Ministries & Local Municipalities:** Who suffer from fragmented communication channels, duplicated administrative overhead across department-specific platforms, reduced public participation, and an inability to track civic needs and policy feedback holistically through a single administrative view.

3. Proposed Solution

Develop a single, trusted mobile application that serves as a unified bridge between citizens and government services in Windhoek and across Namibia. The solution includes the following key operational pillars:

Municipal & Utility Issue Reporting

- Allows citizens to report potholes, malfunctioning traffic lights, burst water pipes, uncollected garbage, and electricity or water supply disruptions.
- Enables photo uploads and precise location capturing (via automatic GPS positioning or a searchable map interface, similar to ride-hailing services like Yango), accompanied by a custom title and detailed description.

Public Policy & Legislative Commentary

- Displays government-published laws and policy proposals fetched directly from the administrative portal.
- Allows users to read proposals, submit written commentary, and provide a star-rating score for specific legislation.
- Provides administrators with a structured backend view displaying user comments alongside the full name of the respondent.

Direct Ministry Submissions

- Enables users to send direct suggestions to specific government bodies (e.g., Ministry of Sport, Youth and National Service; Ministry of Education, Arts and Culture) with support for document attachments.

Offline-First & Local Authentication

- Stores user profile data (full name and phone number) locally on the device.
- Saves reports and submissions locally as drafts during network outages or API downtime, allowing manual or automated submission once connectivity is restored.

Submission History & Tracking

- Features a dedicated history tab enabling users to review sent submissions, preview pending drafts, and verify transmission statuses.

Architecture & Testing Requirements

- Includes pre-populated local dummy data for initial layout testing and user interface verification.
- Follows a clean, maintainable software architecture to ensure long-term stability and scalability.

4. Why It Improves Lives

By eliminating app fatigue and consolidating municipal and civic tools into a single, reliable application, Windhoek residents gain an effortless, transparent channel to highlight municipal damage and actively shape national policy. Eliminating the barrier of multiple app downloads significantly increases public participation and reporting compliance. For local government authorities, a centralized platform streamlines data collection, reduces duplicate submissions, speeds up utility repair turnaround times, and strengthens democratic accountability across Namibia.

5. System Data Model

Mobile App Local Storage Model

Model / Entity	Fields	Purpose
UserProfile	id (UUID), fullName (String), phoneNumber (String), createdAt (Timestamp)	Stores user identity credentials locally on the device.
LocalReport	localId (UUID), serverId (String), category (Enum), title (String), description (String), latitude (Double), longitude (Double), addressText (String),	Manages offline creation, local draft storage, and synchronization tracking for municipal reports.
Model / Entity	Fields	Purpose

	localImagePath (String), syncStatus (Enum: DRAFT, PENDING_SYNC, SYNCED, FAILED), createdAt (Timestamp), syncedAt (Timestamp)	
LocalMinistrySuggestion	localId (UUID), serverId (String), ministryId (String), subject (String), content (String), localDocumentPath (String), syncStatus (Enum), createdAt (Timestamp)	Stores offline direct ministry proposals prior to API synchronization.
CachedLaw	id (String), title (String), summary (String), fullText (String), publishedAt (Timestamp), userRating (Integer), userComment (String)	Caches published legislation locally for offline viewing and draft rating/commentary.

Backend Database Schema

Table Name	Columns & Types	Description
users	id (UUID, PK), full_name (Varchar 150), phone_number (Varchar 20), created_at (Timestamp)	Central registry of unique citizen profiles submitting data.

Table Name	Columns & Types	Description
ministries	id (UUID, PK), name (Varchar 150), code (Varchar 50), contact_email (Varchar 150), is_active (Boolean)	Directory of targeted government entities and line ministries.
reports	id (UUID, PK), user_id (FK), category (Varchar 50), title (Varchar 200), description (Text), location_lat (Decimal), location_lng (Decimal), address_text (Varchar 255), image_url (Varchar 500), status (Enum: RECEIVED, IN_PROGRESS, RESOLVED, REJECTED), created_at (Timestamp), updated_at (Timestamp)	Stores municipal and infrastructure issue submissions.
laws	id (UUID, PK), title (Varchar 255), summary (Text), full_content (Text), document_url (Varchar 500), is_published (Boolean), created_by_admin_id (UUID), published_at (Timestamp)	Published legislative documents and policies open for citizen review.

law_comments	id (UUID, PK), law_id (FK), user_id (FK), rating (SmallInt 1-5), comment_text (Text),	Public ratings and written commentary linked to specific legislation.
Table Name	Columns & Types	Description
	created_at (Timestamp)	
ministry_suggestions	id (UUID, PK), ministry_id (FK), user_id (FK), subject (Varchar 200), content (Text), attachment_url (Varchar 500), status (Enum: NEW, REVIEWED, ARCHIVED), created_at (Timestamp)	Formal citizen proposals submitted directly to specific ministries.

6. Core API & Admin Endpoints

Method	Endpoint	Target Audience	Description
POST	/api/v1/reports	Mobile App	Submit a municipal or utility issue (with image payload and user details).

GET	/api/v1/laws	Mobile App	Fetch active published laws and policy proposals for local caching.
POST	/api/v1/laws/{id}/comments	Mobile App	Submit star ratings and written feedback on a law.

Method	Endpoint	Target Audience	Description
GET	/api/v1/ministries	Mobile App	Retrieve list of active government ministries.
POST	/api/v1/suggestions	Mobile App	Submit direct suggestion with document upload to a target ministry.
GET	/api/v1/admin/reports	Admin Portal	Filter and manage incoming utility issues by category, location, and status.

PATCH	/api/v1/admin/reports/{id}	Admin Portal	Update resolution status of a reported municipal issue.
POST	/api/v1/admin/laws	Admin Portal	Publish new legislative document or policy item for public commentary.
GET	/api/v1/admin/laws/{id}/comments	Admin Portal	View citizen feedback, star ratings, and respondent full names per law.
GET	/api/v1/admin/suggestions	Admin Portal	Access formal citizen proposals
Method	Endpoint	Target Audience	Description
	suggestions		filtered by ministry.